

GPS Master-Slave Clocks With RF Link

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We see public clocks displayed at many locations like office buildings, conference halls, control rooms and so on. Many are just independent clocks displaying Indian Standard Time (IST). The problem is that all the clocks show a slightly different time, none of which may be accurate. Master-slave clocks that are used for tripping analysis are good but are not GPS clocks, whereas GPS clocks are used only in switchyards and other sensitive locations where precise timing is required.

Therefore to get precise time, you either have to rely on the time shown on your Internet-enabled mobile

phone or you have to locate a GPS clock.

Presented here is a set of GPS-enabled master and slave clocks, which show precise time all the time, and the slaves follow the master all



PARTS LIST

Semiconductors:

- IC1 - ATmega328 microcontroller
- IC2 - LM35 temperature sensor
- IC3, IC4 - 7805, 5V regulator
- LED1, LED2 - 5mm LED power indicator
- ZD1 - 3.3V zener diode
- T1 - BC547 npn transistor
- BOARD1 - Arduino Nano board
- DMI - 7-segment display Arduino shield
- LCD1 - 16x2 alphanumeric display
- PZ1, PZ2 - Piezo buzzer
- GPS1 - GPS-2101 (by Rhydolabz)
- TX1 - 433MHz RF transmitter module
- RX1 - 433MHz RF receiver module

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

- R1, R4 - 10-kilo-ohm
- R2 - 22-ohm, 0.5W
- R3, R5 - 680-ohm
- R6 - 100-ohm
- LDR1 - Light dependent resistor
- VR1 - 10-kilo-ohm potentiometer
- VR2 - 10-kilo-ohm preset

Capacitors:

- C1, C2 - 22p ceramic disk
- C3, C5, C8 - 0.1 μ F ceramic disk
- C4, C7 - 10 μ F, 25V electrolytic
- C6 - 100 μ F, 16V electrolytic

Miscellaneous:

- X_{crystal}1 - 16MHz crystal
- ANT.1, ANT.2 - 17cm single-wire antenna
- S1 - Tactile switch
- CON1 - 5-pin connector
- JACK1, JACK2 - DC jack
- 9V DC power supply

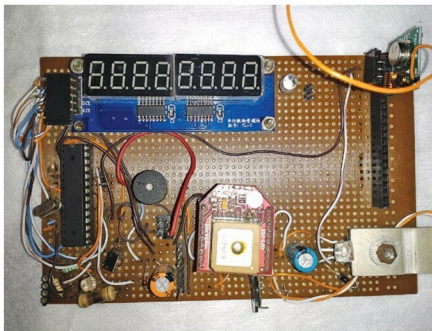


Fig. 1: Author's prototype (GPS master clock)

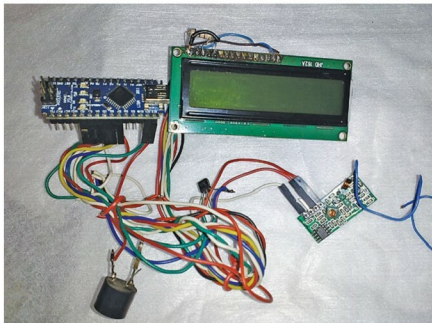


Fig. 2: Author's prototype (GPS slave clock)